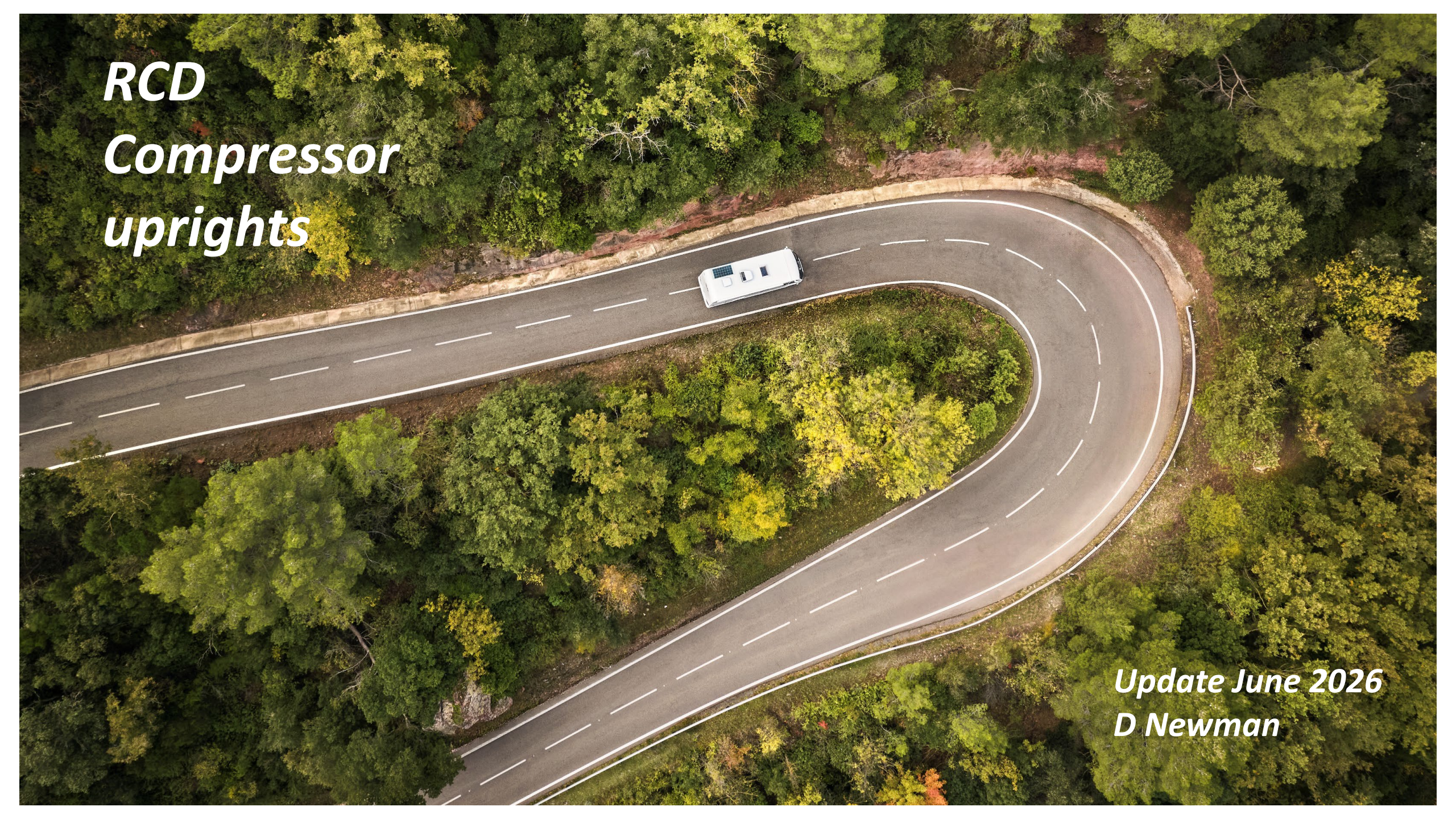


An aerial photograph of a two-lane asphalt road that curves sharply to the right. The road is flanked by dense green trees and foliage. A white van is driving on the road, positioned in the center of the curve. The text 'RCD Compressor uprights' is overlaid in the top left corner.

*RCD
Compressor
uprights*

*Update June 2026
D Newman*



RCD10.5XS (Original Version)

200 Litre upright compressor Fridge - Freezer

Features and benefits:

- Door can be changed - either side open easily
- Food safety compliance of internal components
- User friendly/simple control panel
- Flexible shelf positions to suit users
- Door panel can be removed and customised
- Multiple user modes including "boost" and "quiet"
- Environmentally friendly refrigerant R600a
- Same size/fitment as RMD series
- Fits cavities right back to RM7851 so suitable as direct compressor replacement for 15 – 20-year-old models





RCD10.5XS (XES DH Version)
SKU 9630000157

200 Litre upright compressor Fridge - Freezer

Benefits with Dometic RCD10.5XES DH

* Same as original but with dual opening doors



DOMETIC RCD10.5XES COMPRESSOR REFRIGERATOR 200 L

Designed for smarter use of space, the Dometic RCD10.5XES is powered by compressor cooling technology featuring boost and quiet cooling modes. It comes with an advanced smart door that can adapt to different layouts allowing you to store food and drinks conveniently. The refrigerator comes with a versatile interior and a customisable decor panel that can change to suit all your needs, perfect for your travels.

MAXIMUM STORAGE

FLEXIBLE SET UP

FLEXIBLE INSTALLATION



Specifications

SKU	9600050531
Total Volume	200 Litres
Fridge/Freezer Volume	159/41
Rated Current (12/24/240V)	9.5/4.6/0.5 Amps*
Dimensions (inc Flange WxDxH)	550x615x1260
Weight	48.5KG
Refrigerator type	Compressor
Refrigerant	R600a** (34g)
Temperature range (PC)	0°C to +8C
Temperature Range (FZ)	-20°C to -10°C
Input voltages	12/24 DC 100 – 240 AC

*** This is rated current NOT average consumption!**

**** Butane – volatile but small charge only
(we recommend you don't have a durrty while brazing)**

DOMETIC RCD10.5XES **COMPRESSOR REFRIGERATOR** **200 L DH**

Specifications	
SKU	9630000157
Total Volume	200 Litres
Fridge/Freezer Volume	159/41
Rated Current (12/24/240V)	9.5/4.6/0.5 Amps*
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Installation as shown in Manual

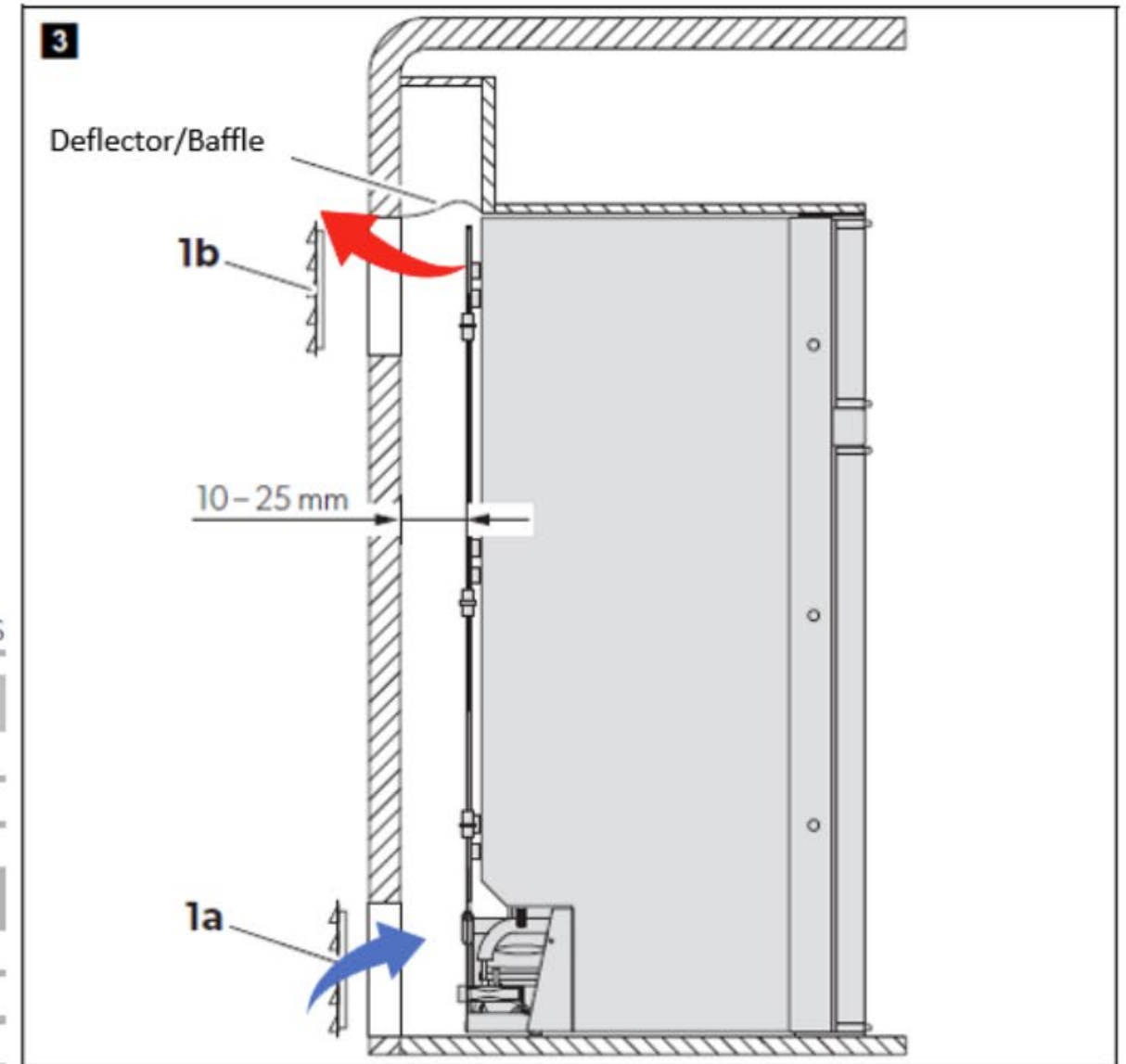
Bottom mounted compressor means you need a lower vent so that cool air can pass over the compressor and electronics.
(Static condenser – fan assisted)

Top vent could also be on the roof of the RV.

Installation

RCD10.5XES

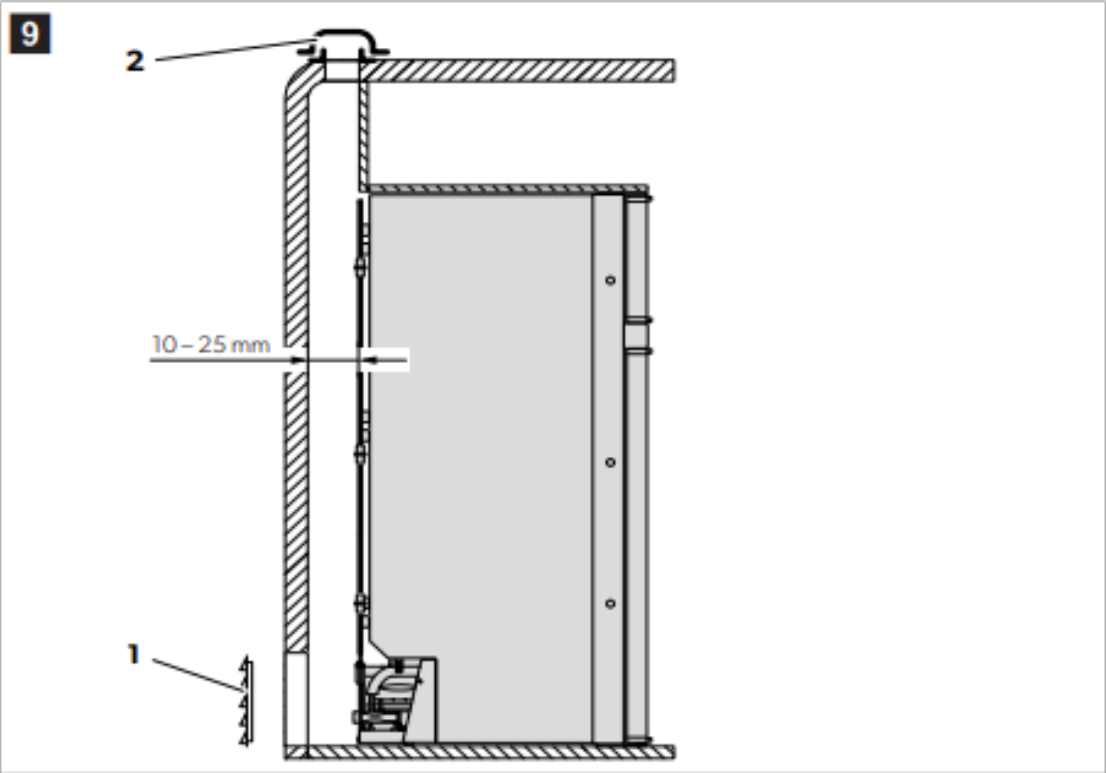
Recess	Overall dimension in mm
Height (A)	1260
Width (B)	550
Depth (C)	615
Recess	Dimension in mm
Height (H)	1246 – 1251
Width (W)	526 – 531
Depth (D)	571 – 586



Some ambiguity from the manual:

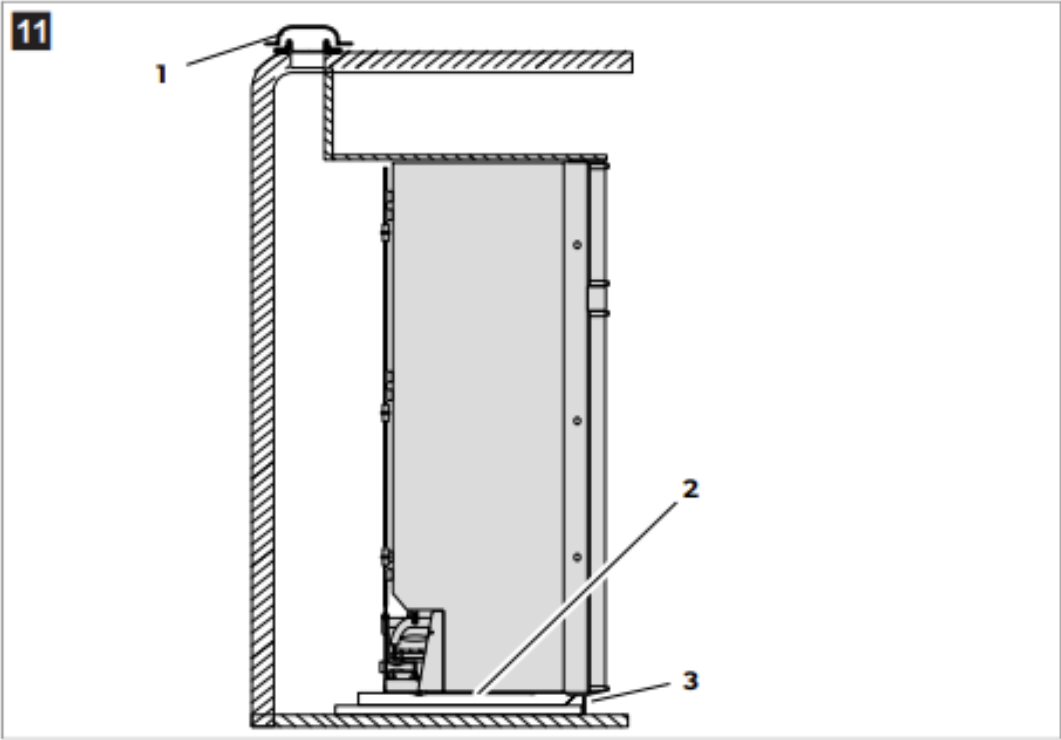
10.2.2 shows how a roof vent should be fitted.
10.4 is less ideal as heat can be trapped
around the back of the freezer area.

10.2.2 Installation with one lower rear vent and one roof vent



Air in from external and out through roof

10.4 Installation with roof ventilation



No. in ☑ Fig. 11 on page 20	Description
1	Roof vent (accessories)
2	Guide rails (accessories)
3	Bottom grille (accessories)

1. Mount the guide rails to the floor (see Mounting the guide rails on page 24).
2. Install the refrigerator and ensure that it is protected from excessive heat, otherwise it leads to poor performance and high power consumption of the refrigerator.
3. Mount the bottom grille (see Mounting the bottom grille on page 25).
4. Install a roof vent. (see Installing the roof vent on page 22)

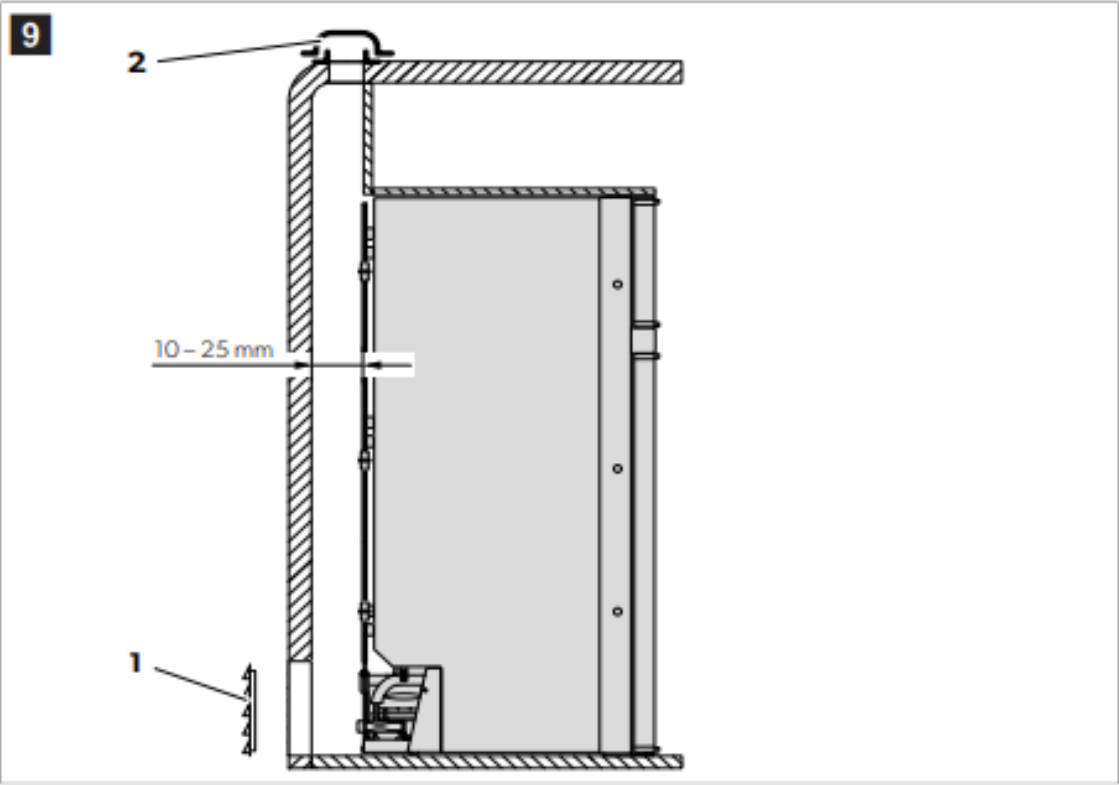
Air in from internal and out through roof

How I believe it should look if fitting a roof vent!

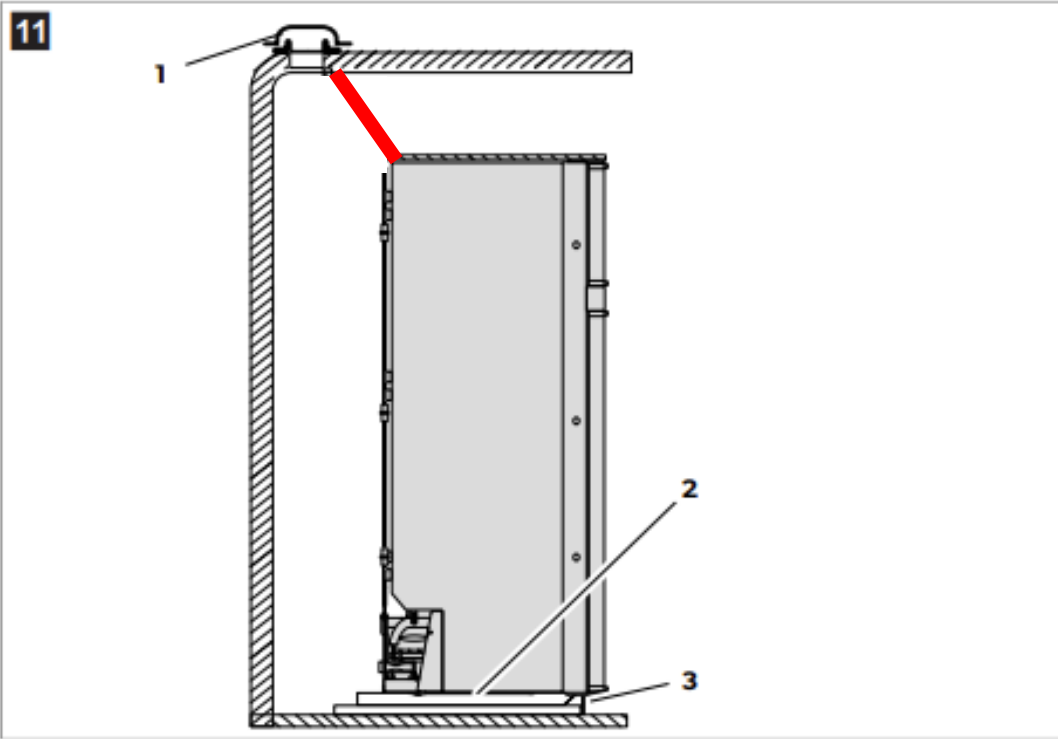
As a BTW – a roof vent for top and internal vent for bottom intake will be more effective from a performance perspective than both top and bottom internal vents – also preventing heat build up inside the van.

- Having no lower external vent can impact on servicing since all external electricals are at the bottom of the refrigerator

10.2.2 Installation with one lower rear vent and one roof vent



10.4 Installation with roof ventilation



No. in ☐ Fig. 11 on page 20	Description
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Venting internally

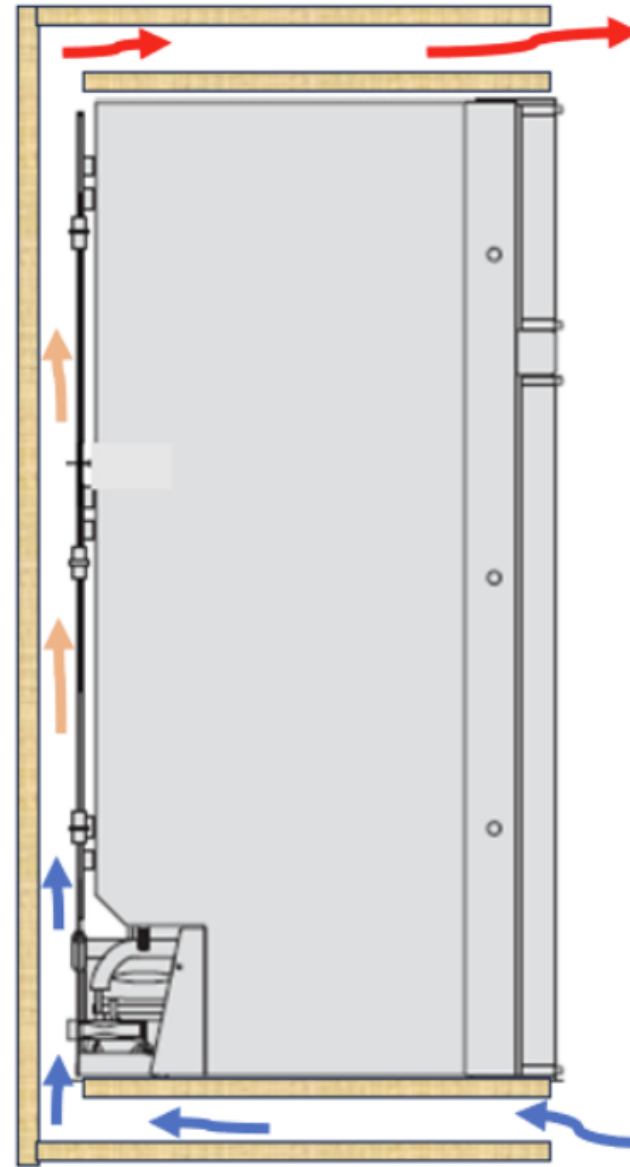
Provided there is an intake vent beneath the refrigerator and an exhaust above the refrigerator, the refrigerator will usually work OK but potentially with slightly lower performance than an externally vented installation.

This condenser is static in nature, but is fan assisted.

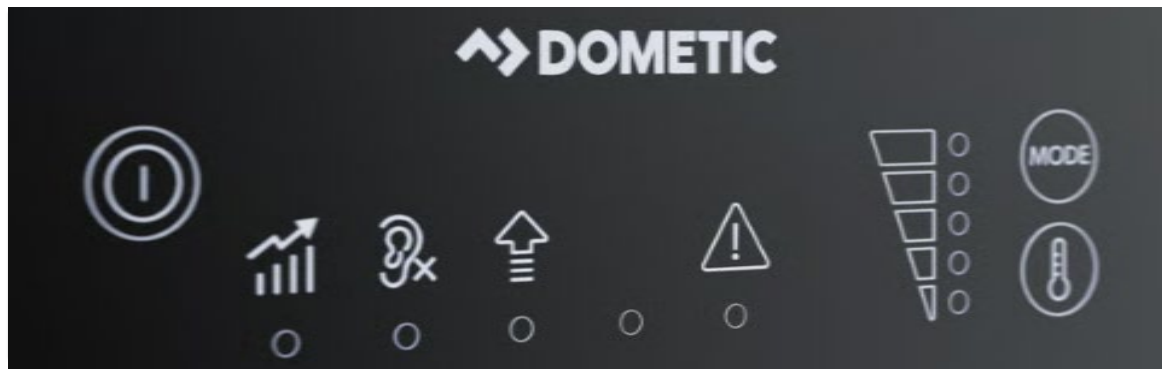
Ventilation always needs a pathway in and out!

Note: If installed without an external lower vent – servicing of the compressor, the control module/dc adaptor and fan will require the refrigerator to be removed from the cavity. If the intention is to limit dust ingress it would be smart to at least fit a closeable service hatch.

- Internal venting kit now available



☐	9620017015 PA01	INTERNAL VENT KIT FOR RUC DH
☐	9620018441 PA01	INTERNAL VENT KIT FOR RUC/RCD



The LED display features the following operating and display elements:

Symbol	Description
	ON/OFF button
	Performance mode
	Silent mode
	Turbo mode
	Error warning
	Temperature indicator
	Mode selection button
	Temperature selection button

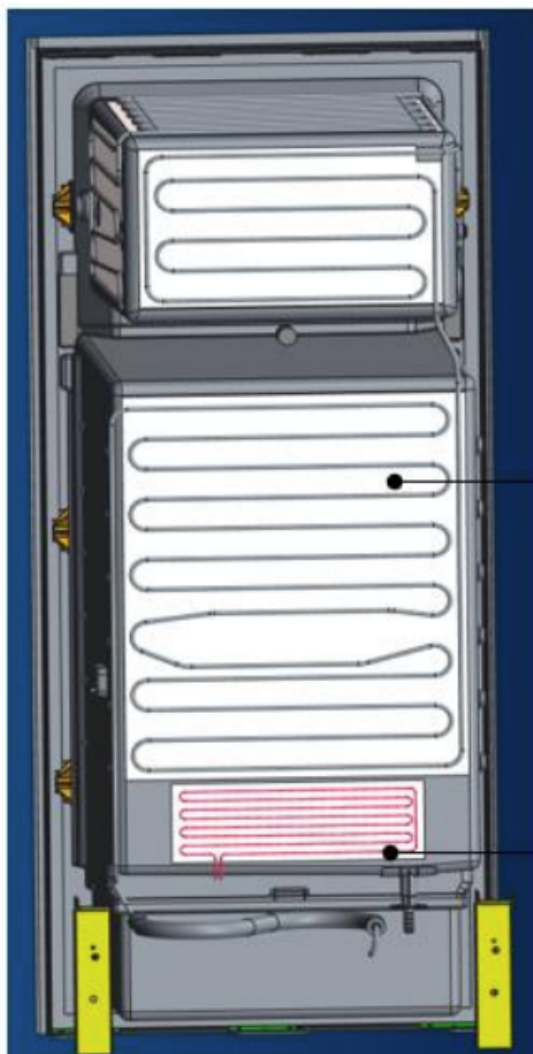
User mode	Symbol	Compressor speed	Outer fan	Defrost	Temperature setting	Duration
Performance		Auto	Auto	Auto	Levels 1 – 5	ongoing
Silent		Limited	Low	Auto	Levels 1 – 3	ongoing
Turbo		High	High	Off	Coldest level (5)	max. 12 hours

Notes:

- Using silent mode in high ambient temperatures will lead to long run times and potentially sub – par performance.
- Selecting Turbo mode in high ambient temperature should be avoided as it needs to get to sub zero temperature (coldest setting) in the PC and may not be able to, although it will revert to normal after 12 hours run time.
- Performance should always be the preferred mode as it automatically adjusts to the prevailing conditions.
- RCD10 & RMD10 fit the cavity for the discontinued RMD8551/5 & RMDX21/25
- **LAC** (Low Ambient Condition) 5 watt heater comes on in off cycle when the ambient temperature is below 12C. This stops the freezer temperature from rising while waiting for fridge to restart

LAC Heater

(Low Ambient Control)



Evaporator

LAC heater



LAC sensor

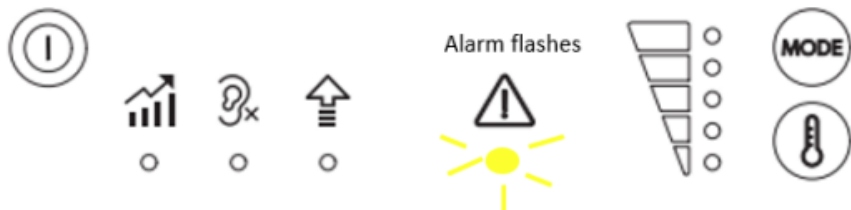
Reason for having such a heater in the PC section:

This style of refrigerator has one temperature sensor controlling the compressor and that is in the refrigerator section; that means that the freezer is a “slave” to the refrigerator. i.e. when the refrigerator section (PC) is cold enough, of course the compressor does not need to run. If the compressor stays off for longer periods in cold ambients, as the PC is already cold enough and only rising slowly, the freezer temperature starts to rise and products that are less dense, such as Ice-cream and Bread, can start to thaw or become softer. Consumer then thinks that there is something wrong, but there isn't – the heater which comes on in the off cycle, ensures that the PC rises in temperature more quickly and forces the compressor to run, which helps the freezer remain at a colder temperature.

How it works in this model: (automatic – not selectable by user)

1. Compressor cycles off
2. LAC sensor realizes that the ambient temperature is below 12°C
3. Heater is turned on – 5 watts is gentle heat
4. When compressor cycles on again the heater is turned off
5. These steps repeat until the LAC sensor realizes 15°C and normal cycling resumes

In case of error malfunction which stops the running of the refrigerator, such as low voltage cut out of the compressor, even below 12C ambient the heater will be turned off



Warning and Error alarms

Flashes	Audio?	Type	Fault	Solution/to check
Solid - on	No	Warning	Defective sensor indication	Replace sensor in PC – requires cut and join as incoming wiring is fixed in vacuum panel. Situated right side of interior of PC behind plastic grill. (Future plug and play solution)
1	Yes	Warning	DC Voltage below 10 or above 31.5	Resets automatically once voltage is restored to normal range. Things to Check: • Wiring & connections • Battery charge and efficiency • Connected equipment such as solar/invertors/adaptors etc • Do they have what they need?
2	Yes	Error	External fan overcurrent	• Reset by pushing on/off button for 2 seconds until a beep is heard • If error appears again replace the fan
3	Yes	Error	Compressor won't start	• Reset by pushing on/off button for 2 seconds until a beep is heard • If error continually appears check: • Power interruptions such as on off with ignition of vehicle causing lockout • Compressor control module • Refrigerant overcharge in a new unit • Locked rotor compressor
4	Yes	Error	Compressor overload	• Reset by pushing on/off button for 2 seconds until a beep is heard • Shut power off for 30 minutes then restart • If large headload recently introduced, reduce the load by half and add later
5	Yes	Error	Compressor overheated	• Reset by pushing on/off button for 2 seconds until a beep is heard • If error remains/returns check; • Ventilation is adequate • External fan is running
6	No	Warning	CI Bus fault	Resets automatically
7	Yes	Warning	Door open too long	After door is open 2 minutes the warning will show and alarm sound – closing the door will reset the alarm. (Does not apply to freezer door)

Built in AC – DC adaptor

- AC voltage is the priority circuit
- DC output when AC applied is 24 volts
- Rated at 135 Watts
- AC to this component first comes through the main PCB
- DC power comes via a terminal block, through the adaptor and back to the main PCB then on to the compressor controller
- When connected to a 12 volt supply it appears possible that you would be feeding 24 volts back to the 12 volts battery the way the wiring is configured, and with a multimeter, you will read 24 volts back at the terminal that your battery is connected to, but there is no real load/capacity available. i.e. you could run a 24 volts 100 milliamp light bulb, but if you were to connect a 24 volts 300 milliamp fan, it would not run. (I tried this) Therefore, it should not be possible to damage 12 volts componentry in line with the wiring input.
- This adaptor could be used to power other 24 volts appliances provided they were < 135 Watts appliances
(Early Waeco dc only refrigerator as one example)

Note: Later adaptors were upgraded to prevent “ghost” voltage. Use 4450036953

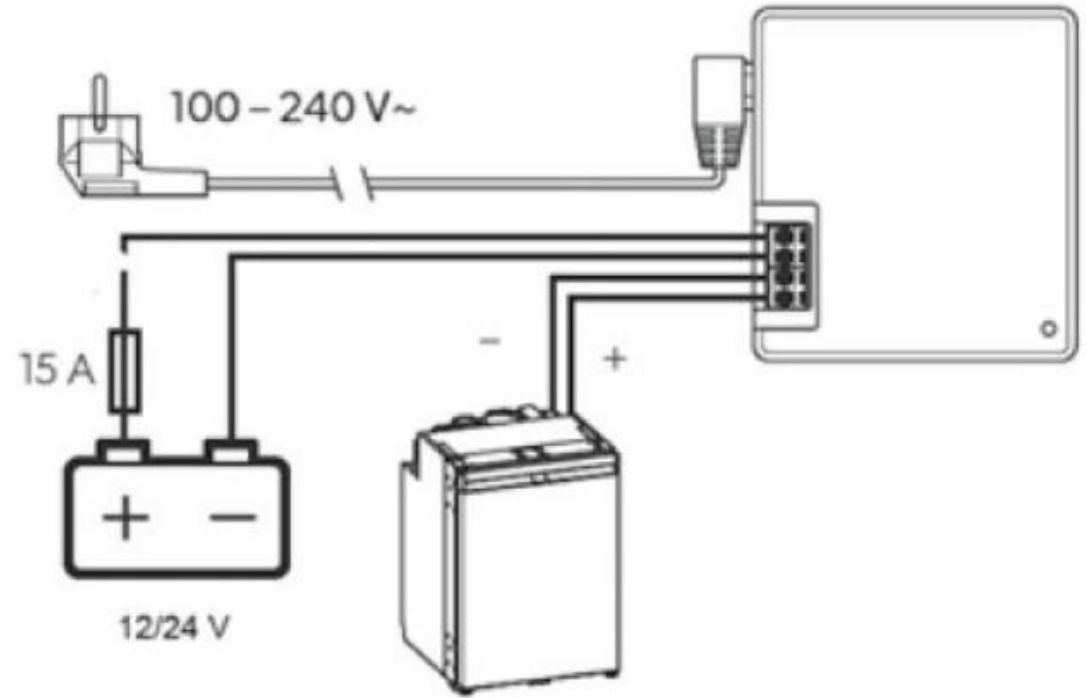


DPS-R135 - AC connection range 110–240 V, it will work with most mains systems in the world.

- Connects 24 V DC compatible refrigerators or other appliances to the mains power socket
- Compatible with all Dometic compressor refrigerators
- Easy quick connect wiring
- Automatic switching DC <> AC (AC Priority)
- Can connect to 12-volt system despite 24-volt output
- Upgrade older refrigerators with Danfoss/Secop or Waeco/Dometic compressors

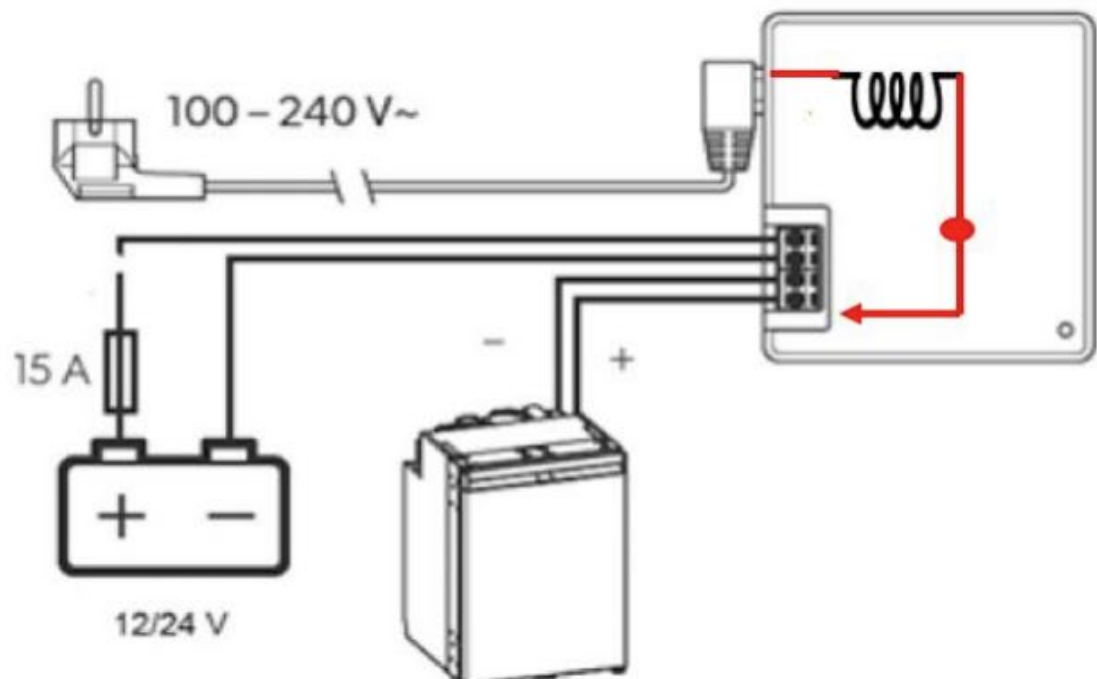


- Similar to AC-MPS50 set up – 12 volt from battery runs fridge and 24 volts from adaptor runs fridge when AC is applied
- (MPS-50 was 150-Watt unit)

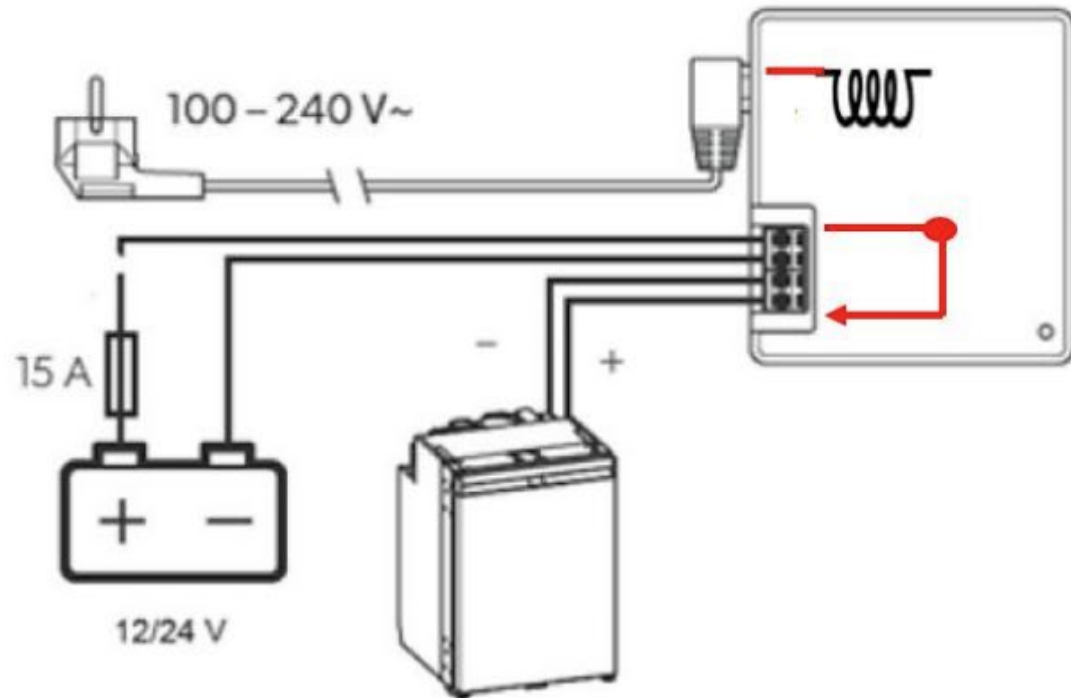


Spares item code now 4450036953 (Suits RCD and NRX models)

24 volts from AC rectified



12 or 24 volts from battery



Feedback may be possible until load is applied – but rare.

Things to be careful/aware of



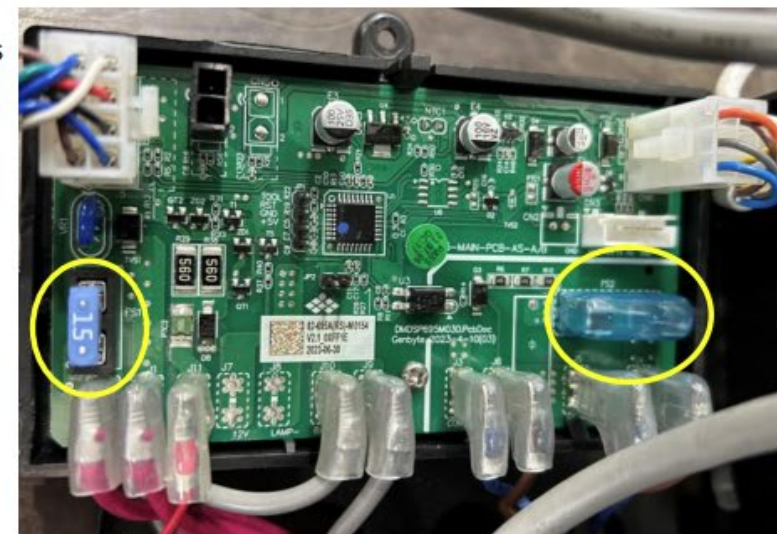
- Do not drill into the sides or rear of refrigerator for any reason; this model has vacuum panels in place to increase the insulation properties. Once you drill into the panel you lose the vacuum

- The condensate drain hose **MUST** be connected or the water will drop onto the fan and potentially cause cabinet damage but also could cause electrical damage/hazard
- Damage as a result of failing to fit or removing the drain hose is unlikely to be seen as warrantable



- Securing screws go through the steel surround so a pilot hole is wise that is of course smaller than the screw to be used. Do not 'over-torque' the driver as you could force the head of the screw through the cabinet if you try hard!
- Make sure your screw is proper head size and length

- Check AC and DC fuses before replacing main or display PCB
- Remember to remove AC power prior to removing cover



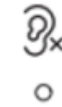
Thermistor/sensor change

("Long Life" thermistor – but things happen!)

Part code for thermistor; 4450034766

Symptoms of a faulty thermistor:

1. Error code shown as constant illuminated alarm LED
2. PC too cold even on warmest setting
3. Compressor not running at all
4. Compressor not cycling off



Alarm flashes



Constant on



1. Thermistor value can be measured at the main PCB at the connection circled. At room temperature the resistance value should be around 12.5K Ohms. If resistance is substantially different or open circuit the thermistor is faulty.



2. Carefully lever off the sensor cover.

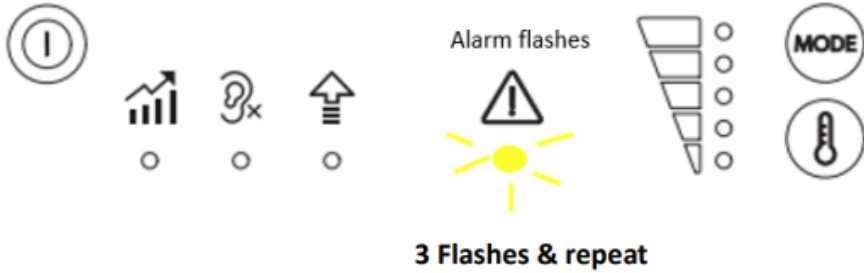


3. Ensure sensor is disconnected from main PCB then cut wiring and strip back insulation leaving enough wiring to solder then heat-shrink the connection.

4. Solder new thermistor in and insulate individual wires with heat shrink as well as the external sheathing to avoid moisture ingress; re-assemble and test.

* Later versions have plug in thermistors.

Compressor control module test



1. Ensure DC power is alive at the top 2 terminals

2. Remove the Orange wire from C and the Blue wire from T terminals

3. Bridge C and T terminals



Symptoms of possible fault:

- 3 Flash error
- Compressor not starting



Terminals are marked on side of unit but a little hard to see



This module is DC 12 or 24 volts only – No AC

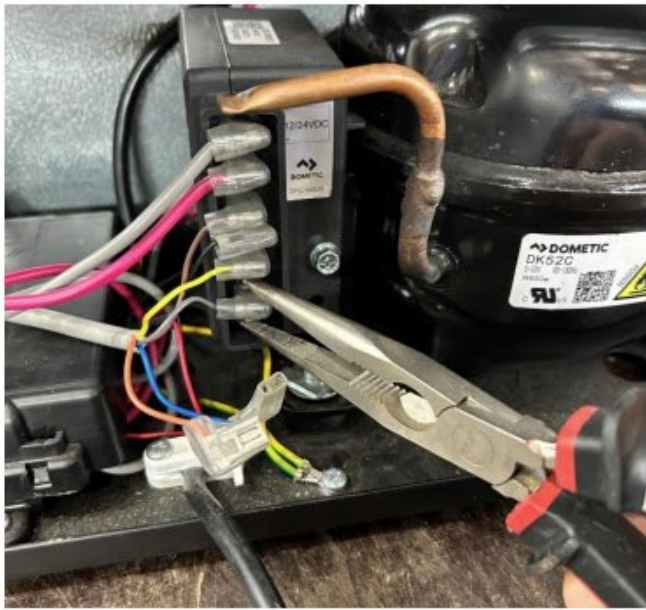
After bridging the terminals give it a few seconds then if the compressor runs, there is nothing wrong with the control unit. (Module)

If compressor does not run, replace the control unit.

If there is definitely power and it still does not run, check the compressor windings for an open circuit situation. Replace compressor if open circuit windings.

Remember that a 3 flash error can also occur as a result of other things such as;

- Refrigerant overcharge leading to high head pressure.
- Constant power interruptions leading to lack of equalization of the refrigerant system. This should automatically reset given enough time.
- A seized compressor – this is highly unlikely though



- It is the terminal you are bridging, not the wires from the terminals.
- No need to make up wiring loop, just pull the wires off and use your pliers to make the bridge for testing.

- Test for resistance value at the 3 pins of the compressor with a multimeter.
- Should have ~ 2.7 ohms across all 3. Open circuit situation means faulty compressor.
- Check that module is properly fitted to the pins – i.e. it hasn't "sprung off" at the back.



Display PCB/Control area



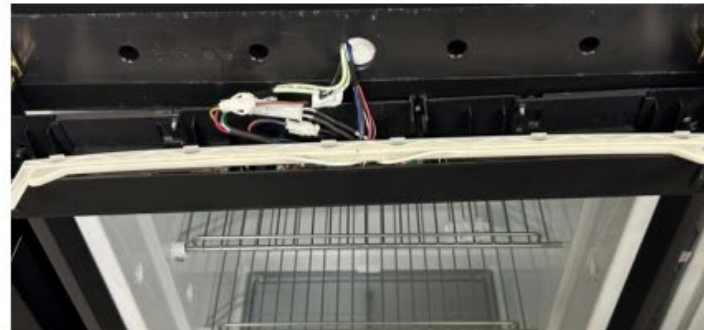
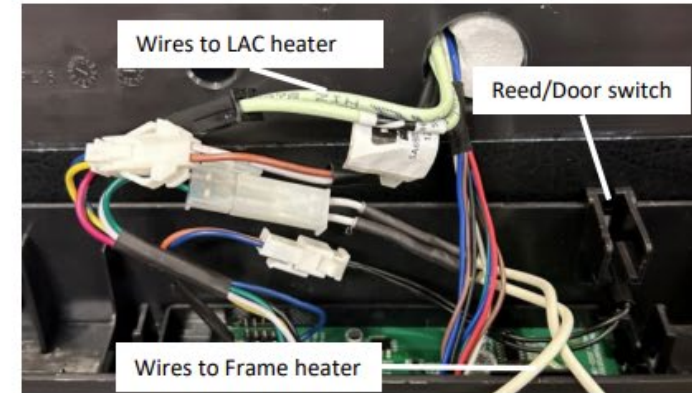
First glance appears that you need to remove the doors to access the Display pcb/control area. You don't, but you do if you need to replace the Frame heater; The PCB and connections can be accessed from the front although can be a little fiddly.



1. With something fine such as a Stanley blade – carefully get under the Decal and peel the whole decal off. Place the decal somewhere safe and clean, sticky side up so that it can be stuck back in place when you finish.



2. With a blade screwdriver push the clips on one end of the PCB mount panel which will release the panel back into the cavity. You can then twist the panel and manoeuvre it diagonally to remove it from the escutcheon – you can then access the PCB and the component connections to test.



Frame heater – this is automatic so not selectable by the user. It warms the freezer divider mullion to prevent condensation. Doors will need to be removed and escutcheon leaned forward if this needs to be replaced

DOORS NOT CLOSING PROPERLY

(DH models)

It is not uncommon to hear complaints of doors not closing properly, doors coming off, not being able to open one side and so on; If the door is closed properly, top and bottom, it will not come off, and you will be able to open from the opposite side next occasion.

These are dual hinged, so it is important that the doors are closed firmly to ensure both sides hinge properly to allow the opposing side to open properly.

People should not be afraid of closing the door firmly, and if they do, they should not have any problem.

The resistance felt as the door bottom gets close to the cabinet is created by the bottom of the door hitting a deliberate ramp to ensure the door is at the correct height thus ensuring proper latching.

See video to right.



Shelves falling out?

Shelves do not fall out, despite popular opinion, “because the cabinet is warped”.

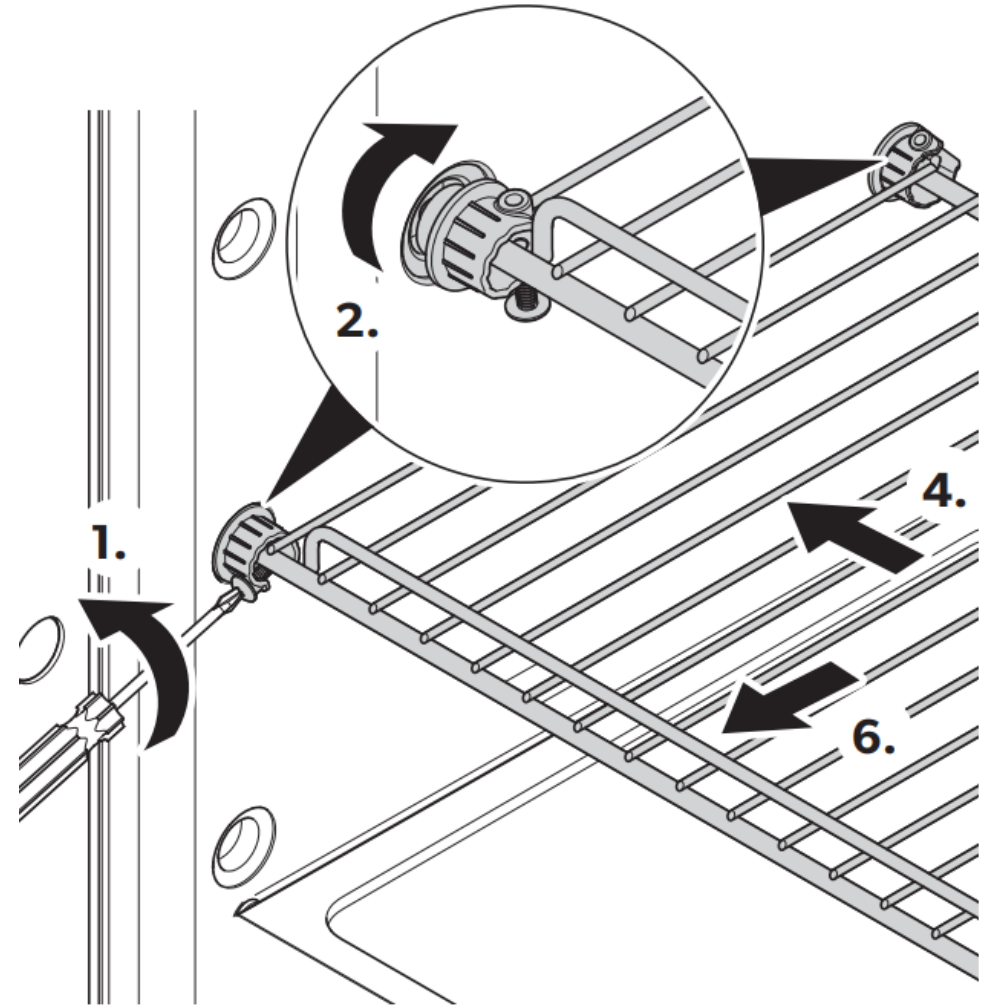
If you think about it, if the shelves supports were longer than the width of the cabinet, how would you ever get them in and out?

Look at the collar.

Loosen the screw, fit the right-side shelf supports in place then lower the shelf on the left to line up with the holes.

Slide the collar into the cabinet receptacle, then turn the collar to a suitable tension point so that the shelf cannot move, then retighten the screw.

This should be checked when hand over of a new van occurs; we have literally had dealers wanting a replacement refrigerator due to a “warped cabinet” when all they needed to do was read and understand the user manual.



Key operating information

Compressor and fan speeds

Mode	Performance		Silent		Turbo	
Level	Compressor Speed	Fan Speed	Compressor Speed	Fan Speed	Compressor Speed	Fan Speed
1	2000	High	2000	Off	N/A	N/A
2	2000	High	2000	Off	N/A	N/A
3	2500	High	2500	Off	N/A	N/A
4	3000	High	N/A	N/A	N/A	N/A
5	3500	High	N/A	N/A	3500	High

Compressor speed can be checked by measuring DC voltage at “C” & “T” terminals of the compressor control module

Voltage	Compressor speed
Zero	2000RPM
One	2500RPM
Two	3000RPM
Three	3500RPM

Fusing in + wire from battery:

12 volt – 15-amp fuse

24 volt - 7.5-amp fuse

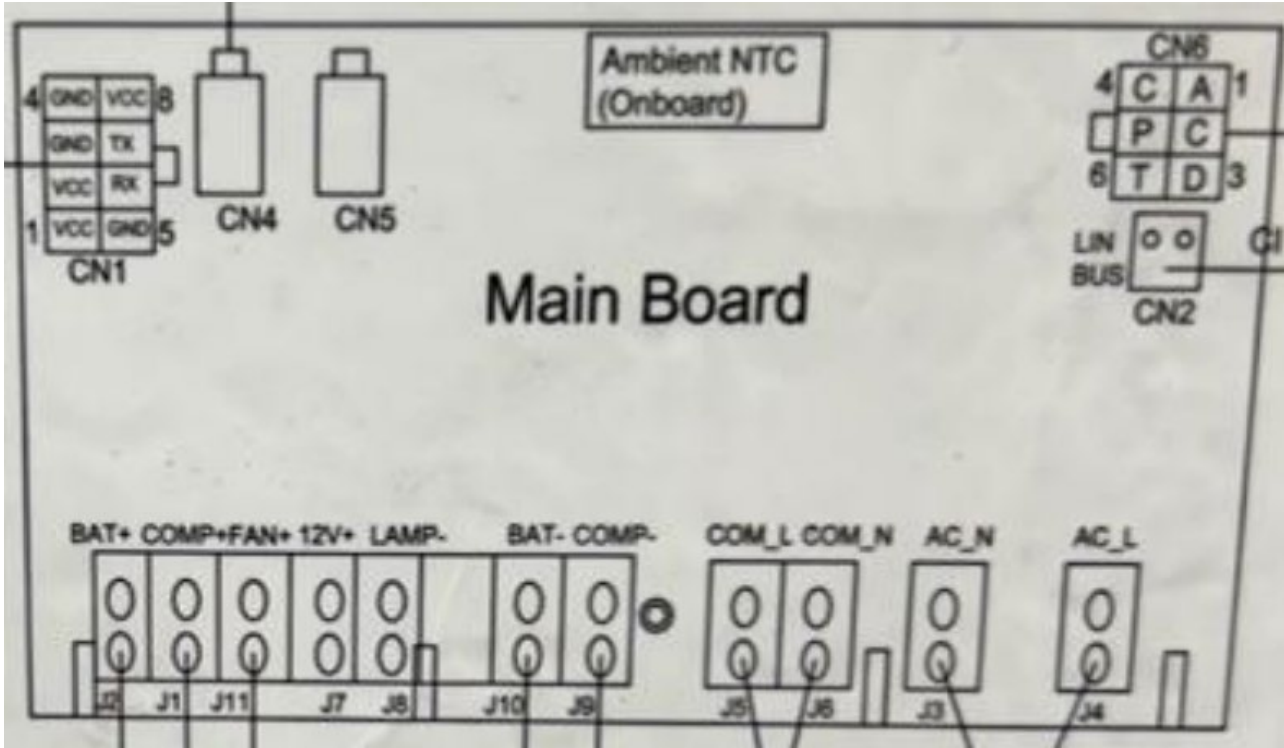
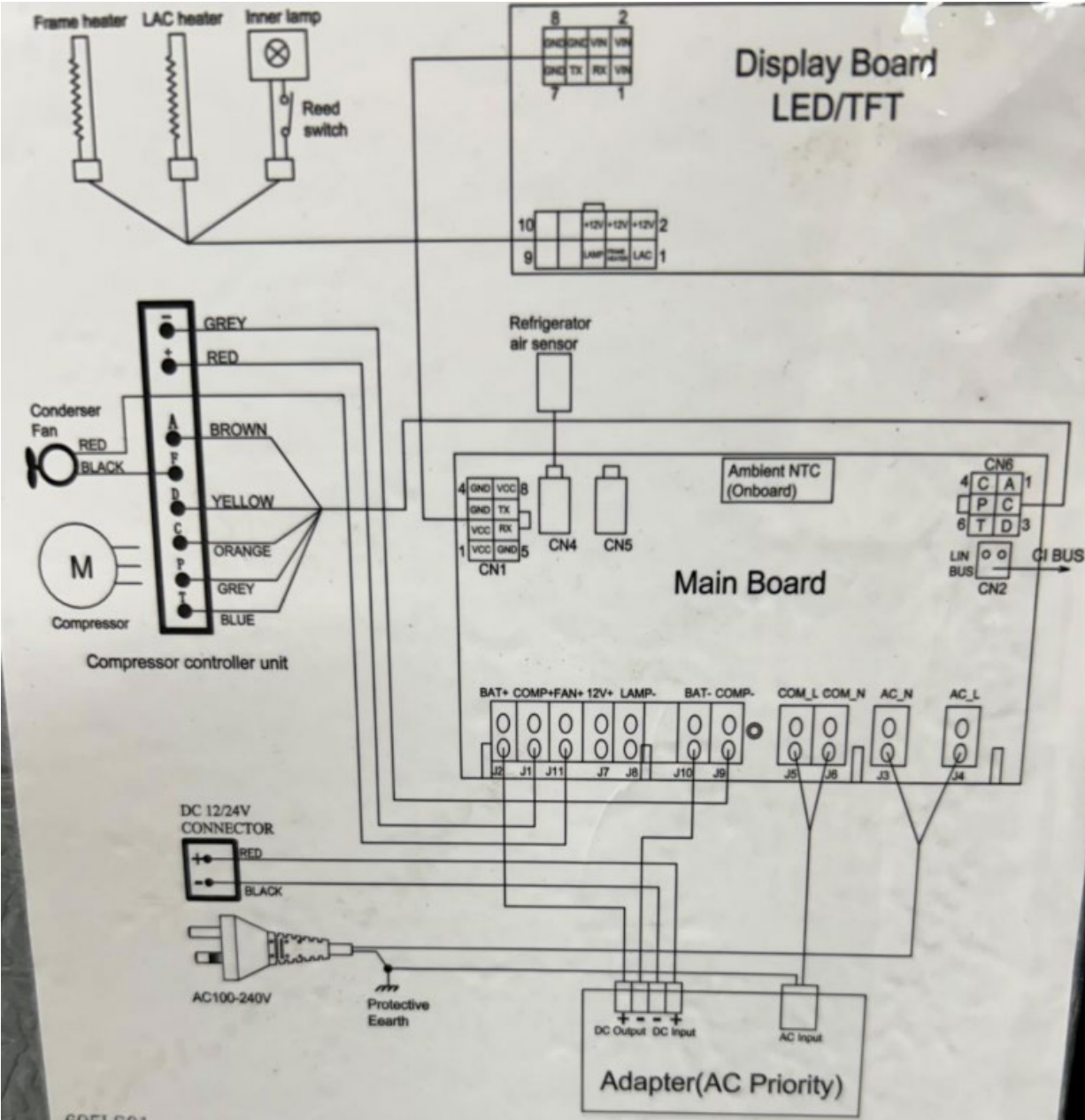
Defrost – Basically after 12 hours accumulative running time, the compressor is forced to run for 60 minutes or to minus 1 C – then the compressor is shut down for one hour. 12 hours resets if power is switched off

Temperature settings*

Level	Cut in	Cut out
1	7°C	6°C
2	5°C	4°C
3	4°C	3°C
4	3°C	2°C
5	2°C	1°C

* Measured at the sensor

Wiring/connection Diagrams



Helpful Information

DC wiring to Refrigerator

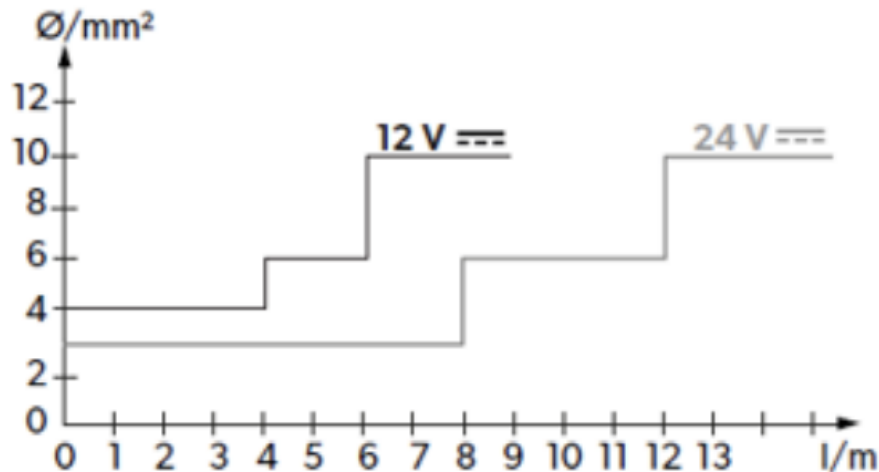
Wiring needs to be sufficiently heavy to carry the load. The thinner the incoming wiring to the refrigerator, the greater the voltage loss.

The longer the run of wiring from the battery to the refrigerator, the greater the voltage loss. All joins/connections along the run are potential voltage loss areas.

Under load, that is as the compressor starts, the voltage loss is increased substantially; without load, that is when the compressor is not running, you will be able to measure what appears to be suitable voltage to enable the refrigerator to run.

The refrigerator compressor can try to start, which causes a voltage loss due to inadequate wiring and the compressor shuts down and a one flash error appears. Immediately the compressor stops, the load is no longer there, so the voltage once again shows higher, and the compressor tries to start again and once again shuts down as the voltage drops. (Over and over)

Don't be fooled, always measure the available voltage under load conditions. Below table is a good guide for wiring to compressor refrigerators.



For extended off grid camping – i.e. no input from alternator



Solar and battery storage

When fitting a solar electrical system, it is important to know what is the likely power drain from the battery storage every day; add up all of the consumers and a good rule of thumb is to try to have around 2/3 in storage.

Example: You expect to consume 100 amphotours each day in total, good practice would be to have 300 Amphotours of battery storage capacity; this helps with cloudy days limiting solar input. It is always better to have more than you need.

What size solar?

With the “electrification” of the industry, much bigger and more elaborate systems are being fitted.

For a basic system, a rough guide/formula to refer could be:

Solar Panel watts, divided by DC voltage, multiplied by 5 sunlight hours per day. SO:

120-Watt panel on a 12 volts system equals 10.

10 multiplied by 5 good sunlight hours equals 50.

Potential provision around 50 Amphotours per day.

So, if you needed 100 Amphotours per day, you would need at least 2 X 120-Watt panels, and it would be better to have more if you have the real estate on the roof of the RV.

Personal opinion, If I want 100 amphotours per day, I am going to fit 300 watts of solar panel & 300 Amphotours of battery. I would go Lithium also as it charges more quickly and holds a higher voltage for longer than AGM/Deep cycle batteries.

Make sure the charger and regulator are MPPT suitable for Lithium.





“Version 3” coming (September - December???)

Some changes to new version:

- *Main PCB gone*
- *LAC gone*



An aerial photograph of a two-lane asphalt road that curves sharply to the right through a dense forest. The road is marked with white dashed lines. A white van is driving on the road, positioned in the upper left quadrant of the frame. The word "END" is written in large, white, italicized capital letters across the center of the road. The surrounding forest is lush with green trees, and a concrete guardrail runs along the outer edge of the curve.

END